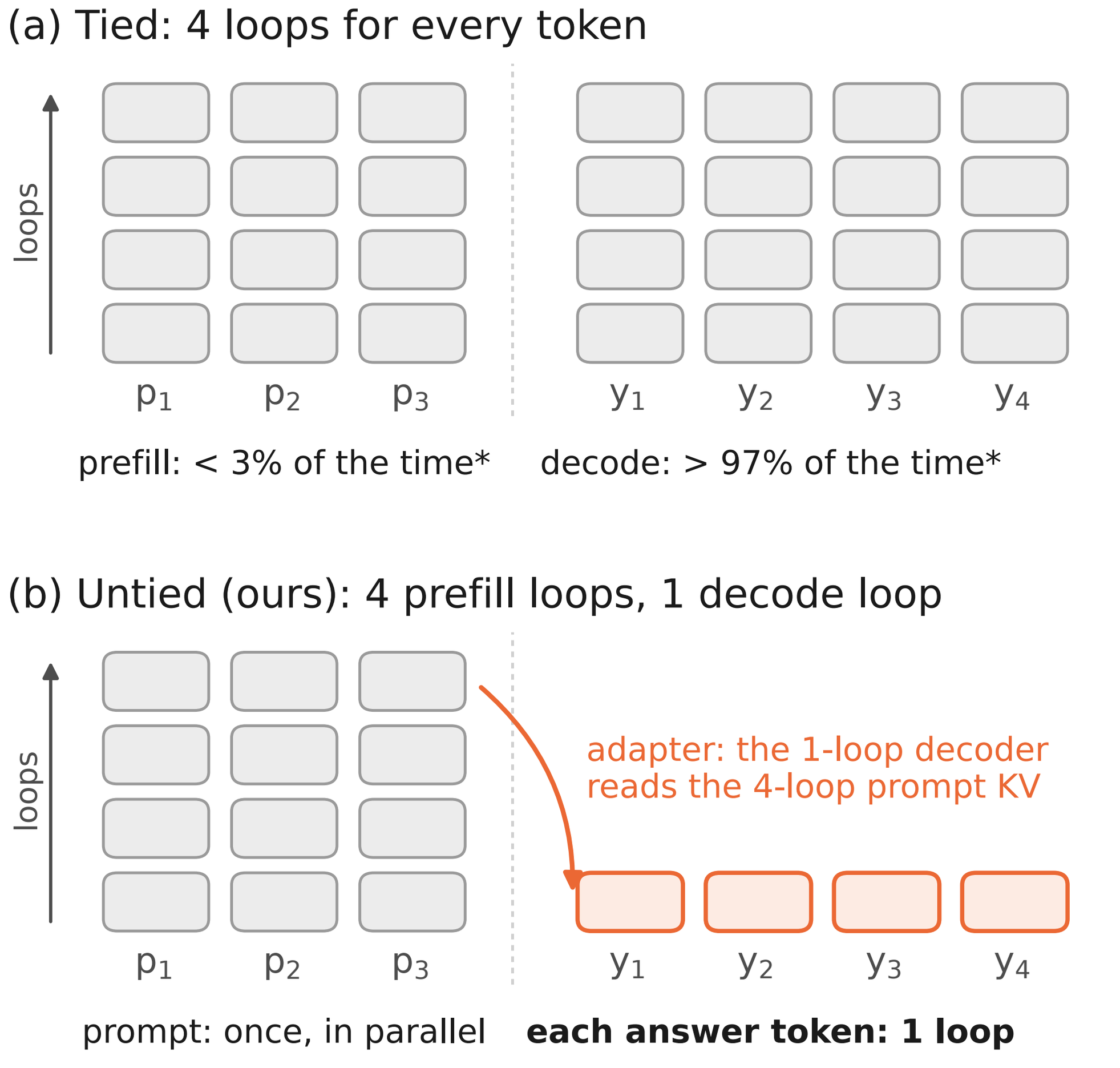
(a) Tied: 4 loops for every token
loops
p1
p2
p3
y1
y2
y3
y4
prefill: < 3% of the time*
decode: > 97% of the time*
(b) Untied (ours): 4 prefill loops, 1 decode loop
loops
adapter: the 1-loop decoder
reads the 4-loop prompt KV
p1
p2
p3
y1
y2
y3
y4
prompt: once, in parallel
each answer token: 1 loop